🔑 **Answer Key — Teacher Copy · Lesson 5-7**

MSTAR PRACTICE WORKSHEET

Lesson 5-7: Determine Surface Area of Prisms

Grade 6 Mathematics · Standard 6.GR.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the surface area of a rectangular prism with l = 5 cm, w = 4 cm, h = 3 cm?

- A 47 cm^2
- B 60 cm^2
- C 94 cm^2
- D 94 cm^3

SA = 2(5×4) + 2(5×3) + 2(4×3) = 40 + 30 + 24 = 94 cm^2.

Question 2 SELECTED RESPONSE — CORRECT: C

How many faces does a triangular prism have?

- A 3
- B 4
- C 5
- D 6

A triangular prism has 5 faces: 2 triangular bases and 3 rectangular lateral faces.

Question 3 SELECTED RESPONSE — CORRECT: C

A triangular prism has two triangular bases each with area 12 cm² and three rectangular faces with areas 30 cm², 40 cm², and 40 cm². What is the total surface area?

A) 110 cm²

B) 122 cm²

C) 134 cm²

D) 134 cm³

> SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 cm².

Question 4 SELECTED RESPONSE — CORRECT: C

A triangular prism has triangular bases with base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What is the surface area?

A) 124 in²

B) 160 in²

C) 184 in²

D) 184 in³

> Triangular bases: 2 × (½ × 6 × 4) = 24 in². Three rectangular faces: 6(10) + 5(10) + 5(10) = 60 + 50 + 50 = 160 in². SA = 24 + 160 = 184 in².

Question 5 SELECTED RESPONSE — CORRECT: C

A rectangular time capsule box measures 3 in by 4 in by 5 in. How much wrapping paper covers its surface (no overlap)?

A) 47 in²

B) 60 in²

C) 94 in²

D) 94 in³

> SA = 2(lw + lh + wh) = 2(3×4 + 3×5 + 4×5) = 2(12 + 15 + 20) = 2(47) = 94 in².

Question 6 SELECTED RESPONSE — CORRECT: C

A rectangular capsule crate is 6 cm long, 5 cm wide, and 4 cm tall. How much cardboard is needed to cover all its faces?

- (A) 74 cm^2
- (B) 120 cm^2
- (C) 148 cm^2
- (D) 148 cm^3

> SA = 2(6×5) + 2(6×4) + 2(5×4) = 60 + 48 + 40 = 148 cm^2.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A rectangular prism measures 6 cm long, 4 cm wide, and 2 cm tall. What is its surface area?

- (A) 24 square centimeters
- (B) 44 square centimeters
- (C) 48 square centimeters
- (D) 88 square centimeters

> 2(6×4) + 2(6×2) + 2(4×2) = 48 + 24 + 16 = 88 square centimeters.

- **A:** That is the VOLUME, 6 times 4 times 2, in cubic units. Surface area totals the six faces in square units.
- **B:** That counts one of each face instead of two. Every face has a matching opposite.
- **C:** That is the two largest faces only, 2 times 6 times 4. Four more faces remain.

Part B — correct answer: A

Why is surface area measured in square units while volume is measured in cubic units?

- (A) Surface area covers the outside faces, which are flat regions; volume fills the inside space.
- (B) Surface area is always smaller than volume.
- (C) Square units are used whenever a shape has straight sides.
- (D) Volume uses cubic units only for prisms.

> Surface area is the total of flat face areas, so it uses square units. Volume measures how much space the solid fills, so it uses cubic units.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Counting Only Three Faces

> Priya found the surface area of a box measuring 5 by 3 by 2 units. She wrote: '5×3 + 5×2 + 3×2 = 15 + 10 + 6 = 31 square units.'

Explain Priya's misconception and give the correct surface area.

> **Model answer:** Surface area = 2(15) + 2(10) + 2(6) = 62 square units. Priya found only three faces; every face has a matching opposite face, so each product is doubled.

2 points	Student identifies that a rectangular prism has 6 faces in 3 matching pairs, so each product must be doubled, and gives 62 square units.
1 point	Student gives 62 square units without explaining the matching pairs.
0 points	Student agrees with Priya or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A company ships products in two container options: a rectangular prism (8 × 6 × 4 in) or a triangular prism (triangle base 8 in, triangle height 6 in, prism length 8 in, slant sides 6.3 in each). Both must be wrapped in protective film. Which shape uses LESS film? Show your work.

> **Model answer:** Film covers the outside, so compare surface areas. The rectangular prism gives 2(8×6) + 2(8×4) + 2(6×4) = 96 + 64 + 48 = 208 in², while the triangular prism gives two triangles 2(½ × 8 × 6) = 48 plus three rectangles 8×8 = 64 and two of 6.3×8 = 50.4, for 212.8 in². The rectangular container uses less film, by about 4.8 in².

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.